



Outlook on the global Logging and Tracing Solution

MICROSAR Classic and MICROSAR Adaptive

Agenda

► Introduction

Product Overview Logging

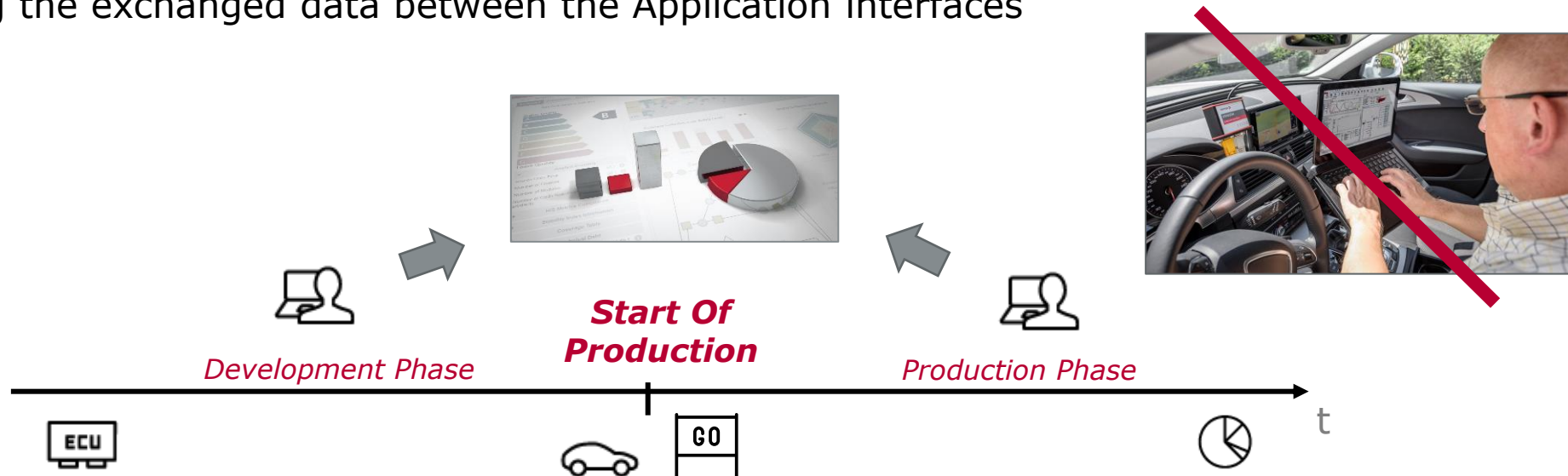
Product Overview Tracing

Product Overview IPC Measurement

Vehicle Log & Trace

Motivation

- ▶ **Analysing and testing Vehicles becomes a more relevant challenge**
 - ▶ More software is deployed
 - ▶ More computation power is necessary – HPCs
 - ▶ Development cycle exceeds the Start of Production (SOP)
- ▶ **The Vector MICROSAR Vehicle Log & Trace Solution tries to provide the necessary solutions**
 - ▶ Logging of the system's runtime behavior
 - ▶ Tracing the execution of the program flow
 - ▶ Measuring the exchanged data between the Application interfaces



Agenda

Introduction

► **Product Overview Logging**

Product Overview Tracing

Product Overview IPC Measurement

Vehicle Log & Trace

MICROSAR Classic – Diagnostic Log And Trace

► Diagnostic Log And Trace Module

- Support of XCP Protocol
- Support of Dlt Protocol

► Application APIs:

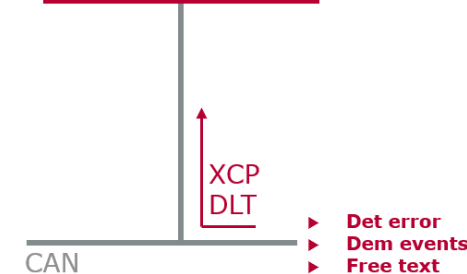
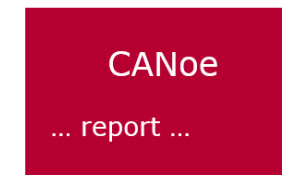
- Provides a Verbose Logging API
- Provides a non-Verbose Logging API
- Provides a Tracing API

► BSW Events:

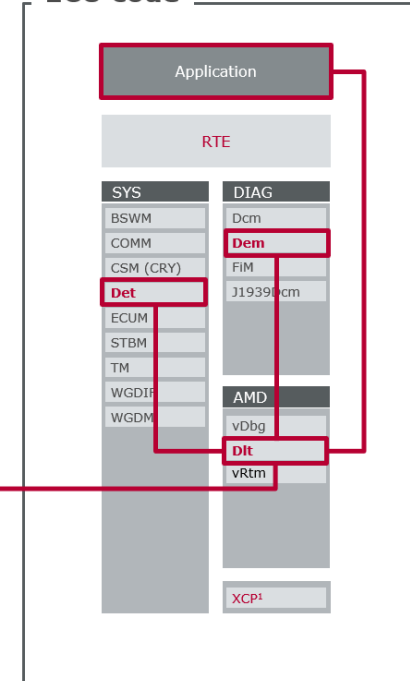
- Forwards Det Events
- Forwards Dem Events, with and without Dtc

AMD **Dlt on Xcp**
AMD **Dlt on PduR**

Diagnostic Log and Trace



ECU code



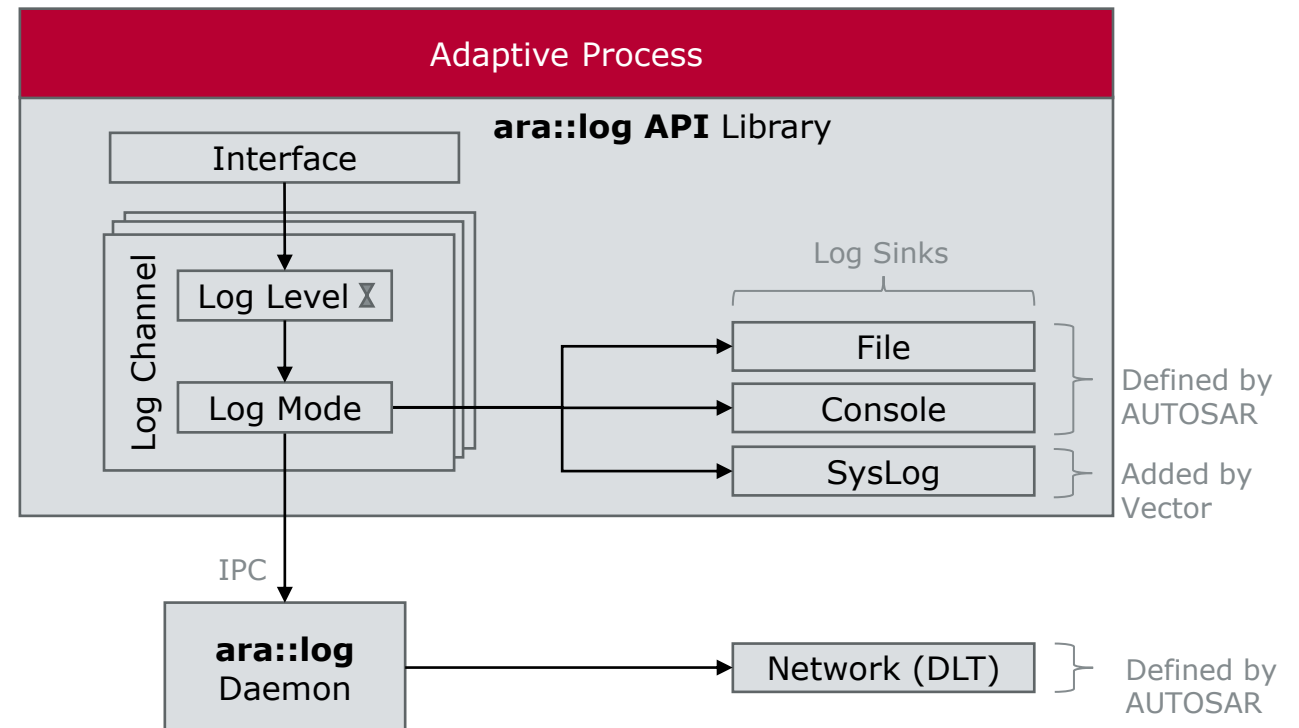
MICROSAR Adaptive – ara::log API Library

► Log And Trace Library

- Library for Applications to log with different LogLevels and Contexts
- Provides a Verbose API
- Configuration of LogLevels via JSON config file

► Log Sinks:

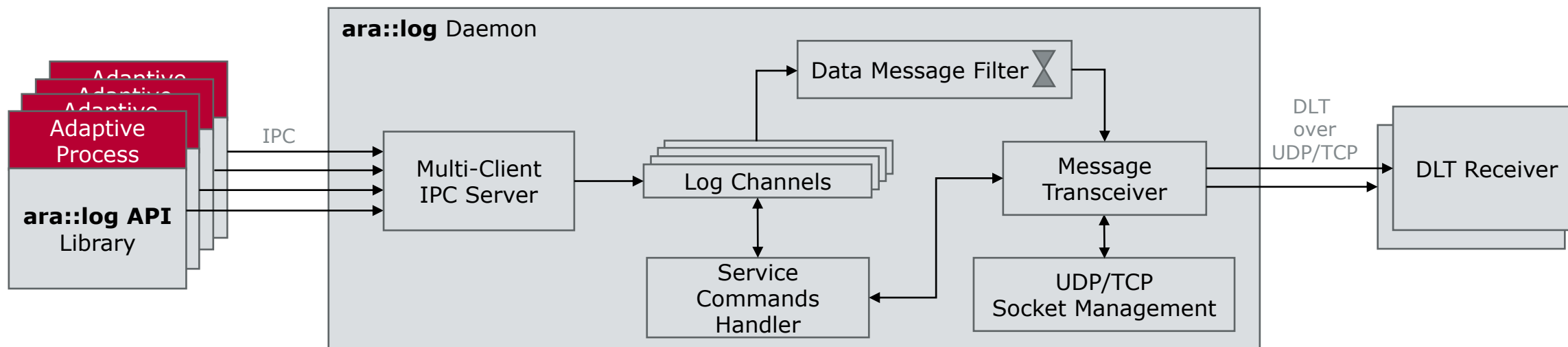
- Support of Dlt Protocol (Remote)
- Support of File Logging
- Support of Console Logging
- Supports logging to SysLog



MICROSAR Adaptive – ara::log Daemon

► Log Daemon

- Aggregates logs from the Applications
- Serializes Dlt Protocol
- Compatible to GENIVI Dlt Viewer
- Supports UDP and TCP messageing
- **Planned: Supports receiving and dispatching Dlt Service Commands**



Agenda

Introduction

Product Overview Logging

► **Product Overview Tracing**

Product Overview IPC Measurement

Vehicle Log & Trace

MICROSAR Adaptive ara::com Tracing

Performance Trace-Points

- ▶ Ara::com implements C++ Comments which mark program execution relevant code points
- ▶ These comments are called PTP – Performance Trace Points
- ▶ The PTP can be replaced by a script with each possible tracing mechanism for example:
 - ▶ USDT Trace Macros
 - ▶ QNX trace event API
- ▶ Each Comment follows a certain naming scheme:

```
// PTP-B-IpcBinding-ActiveConnection_Send  
performanceTracepoint-Begin-<NameOfBinding>-<NameOfTracePoint>
```

← DTRACE_PROBE(provider, name, arg);

```
// PTP-E-IpcBinding-ActiveConnection_Send  
performanceTracepoint-End-<NameOfBinding>-<NameOfTracePoint>
```

For Development Only

- ▶ Manipulates safety qualified code
- ▶ Generates huge amount of data

Agenda

Introduction

Product Overview Logging

Product Overview Tracing

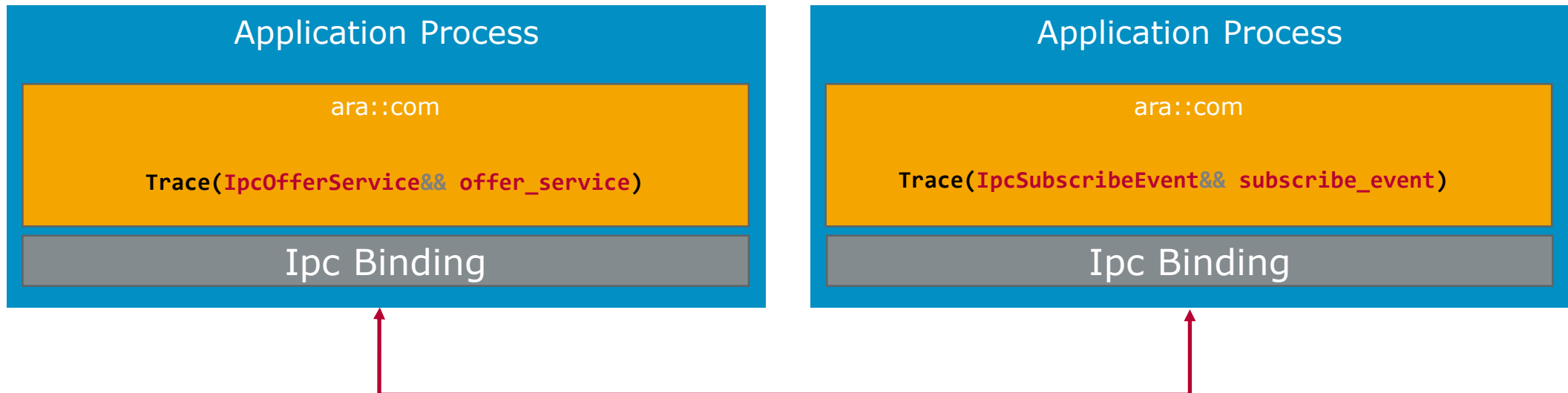
► **Product Overview IPC Measurement**

Vehicle Log & Trace

MICROSAR Adaptive ara::com IPC Measurement

Measurement of IPC communication via ara::com

- ▶ The communication between distributed software in the AUTOSAR Classic world is measurable on the bus
- ▶ Integrated software in one High Performance Computer exchanges data within the controller (IPC)
- ▶ This data is typically very hard to be measured
- ▶ Ara::com provides "Trace Interfaces" (Callouts) for IPC communication to access the IPC Payload on measurement related events (amsr::ipc_binding::TraceIf)



Agenda

Introduction

Product Overview Logging

Product Overview Tracing

Product Overview IPC Measurement

► **Vehicle Log & Trace**

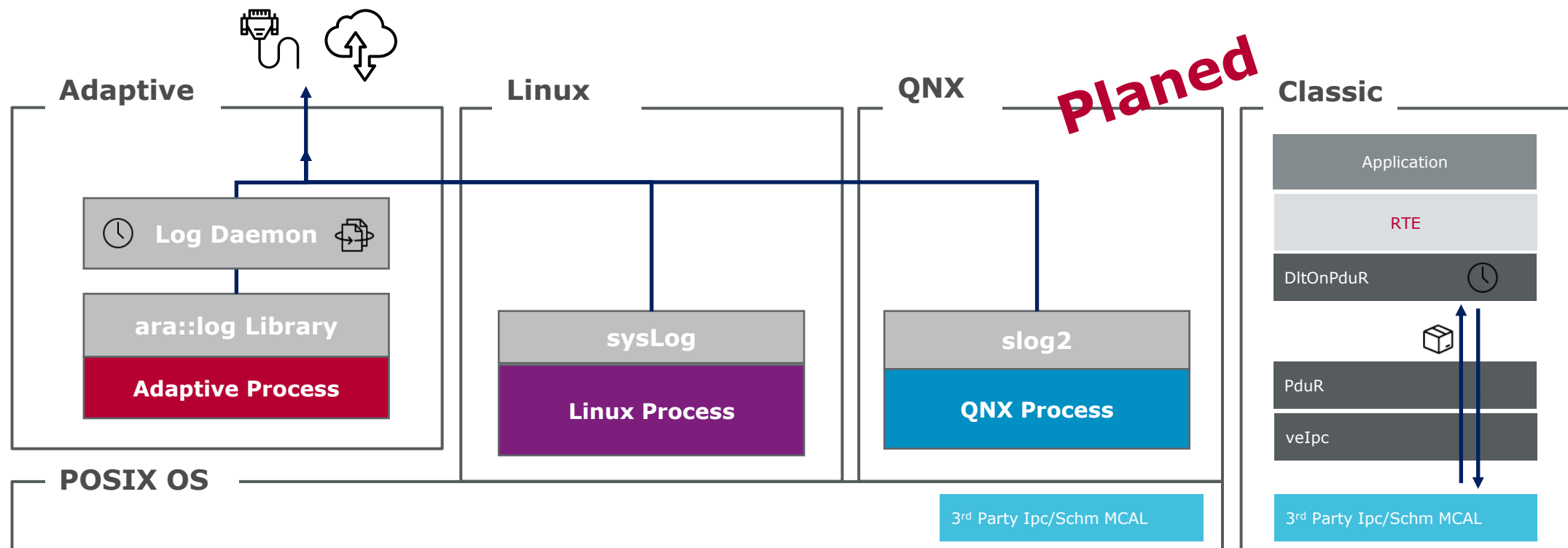
Overview

Aggregation of Log&Trace messages throughout the Vehicle

- ▶ Collection of Linux SysLog messages
- ▶ Collection of QNX Slogger2 messages
- ▶ Collection of AUTOSAR Classic Dlt PDUs

Possible Use-Cases

- ▶ System Analysis – Local Connection
- ▶ Vehicle and Fleet analysis – Cloud Connection



For more information about Vector
and our products please visit

www.vector.com

Authors:
Ziegler, Sebastian
Vector Germany

